and cannot establish about urban noise

Hanoi, 363 field measurements, three negative results
and the instrument we ended up building

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Outline

Everything on these slides is reproducible from `github.com/Colauz/noise-modelling-hanoi`

Context and research questions

The gap this project sits in

- Hanoi: dense, fast-growing, **motorcycle-dominated** traffic
- **No national strategic noise-mapping programme** and no statutory noise action plan in Vietnam
  Nguyen et al. 2025, *City Environ. Interactions*
- **One** instrumented campaign ever published for Hanoi — and it measured in **2005–2007**
  Phan et al. 2010, *Applied Acoustics* 71(5)
- Class 1 sound level meters and commercial propagation software: out of reach for most local research budgets

The question

How far does a protocol built from **three consumer smartphones** and **open data** actually get you?

The honest answer turned out to be: further than nothing, and much less far than a map.

Research questions — and what is out of scope

Four questions, fixed before the analysis

1. Can urban morphology from OpenStreetMap predict measured noise at an **unmeasured location**?
2. Does a model trained in one city **transfer** to another?
3. Does vehicle **flow extracted from video** explain measured levels?
4. What can a smartphone protocol claim with **no absolute reference**?

Declared out of scope — stated up front, not discovered later

- Night-time noise: 10 measurements between 21:00 and 06:00, none between 00:00 and 05:00
- Compliance assessment against any standard
- Any prediction outside the three measured sites

Result preview

Q1 — weakly, $R^2 \approx 0.25$

Q2 — **no**

Q3 — **no**

Q4 — contrasts, never levels

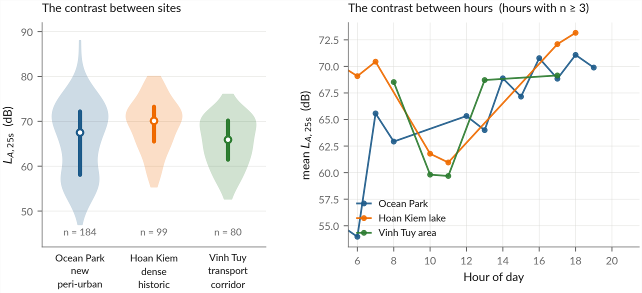
Related work — ten references, selected by what they are used for

Starting point	Method	Anchoring
Sunbird Urban Noise Uganda 61K — 61k clips, *Sci. Data* 2026	Roberts et al. 2017, *Ecography* — spatial CV must exclude the feature support	Phan et al. 2010 — the only instrumented Hanoi campaign
The dataset we set out to transfer from	The paper that told us our first R^2 was wrong	The reference our levels are compared against
Also used: CNOSSOS-EU · NoiseModelling · GAMA · Gelb & Apparicio 2019 (dosimeters, HCMC) · QCVN 26:2010/BTNMT · TCVN 7878-2:2010		

Full annotated bibliography with verified DOIs: `docs/literature-review.md`

Data and protocol

The campaign



`data/processed/measurements.csv`

Protocol — 3 handsets, *Decibel X* (A-weighted, SLOW), 20–30 s per point, a clip $\geq$ 10 s, a 10 m GPS gate, ODK Collect $\rightarrow$ Kobo.

Published, CC-BY-4.0 — both CSVs and both XLSForms. Videos and raw Kobo exports are **not**: faces, plates, collector identities.

Metrology — the constraint everything else is built around

The three phones were cross-calibrated against *each other*, never against a reference instrument

A bias common to all three is **invisible in the data by construction**.

What that forbids

- Stating an absolute level as a fact
- Any compliance claim under QCVN 26:2010 — which requires a class 1 or class 2 instrument
- Comparing a 25 s sample to an annual L_{den}

What it still allows

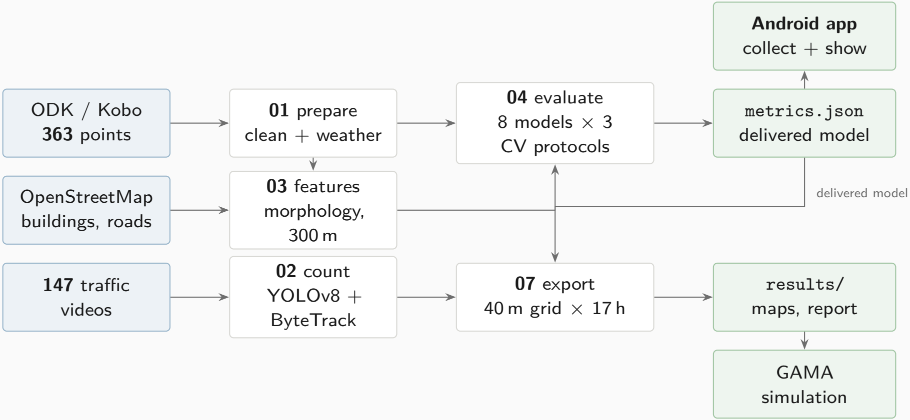
- Contrasts *between places* — bias cancels in a difference
- Contrasts *between hours*
- A *bounded* absolute-bias interval, from instrumented literature: $+3.0$ **to** $+7.3\,\mathrm{dB}$

The target quantity is $L_{A,25\mathrm{s}}$: a 20–30 s A-weighted sample. **Not** L_{Aeq}, L_{den} or L_{night}.

`docs/metrology.md` · `results/tables/literature_anchoring.md` the interval is estimated, never applied

Method and evaluation protocol

The chain, end to end



The delivered model is chosen **by code** — nothing downstream can silently substitute another.

`scripts/01_...11_` · `docs/methodology.md`

The evaluation protocol — and why it is the whole story

Every model is scored under **three splits**, on exactly the same folds:

1. **Block-CV, 600 m** — square blocks in UTM 48N, `GroupKFold` on the block. The permissive one.
2. **Buffered leave-one-out, 300 m** — for each point, train on *all* points more than 300 m away.
   The reference protocol.
3. **Leave-one-site-out** — generalisation to an unseen urban typology.

Every score carries a 95 % interval bootstrapped **by block**, not by point: resampling correlated points gives an interval that is far too narrow.

Why 300 m and not less

The morphology features aggregate over a disc of **radius 300 m**. Two points 110 m apart share more than **85 %** of that disc.

Group on 110 m cells and the model sees near-twins of its test points in training. The score is then not out-of-sample.

Roberts et al. 2017, *Ecography*

`scripts/04_evaluate_models.py`

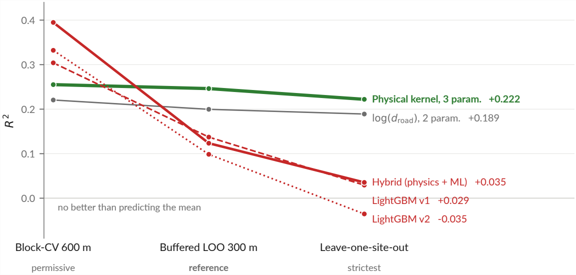
Eight models, three baselines, one ablation

Nothing is compared against nothing. The baselines are the point:

Baselines — what any model must beat to have said anything	
`global_mean`	the absolute floor
`site_mean`	no fine spatial information at all
`site_hour_mean`	mean per (site, hour) — **the one to beat**
`dist_road`	linear regression on $\log(d_{\mathrm{road}})$ — minimal physics
`idw`	inverse distance weighting — pure interpolation
Ablation — which half of the feature set is doing the work	
`lgbm_time`	hour + weekend only
`lgbm_morpho`	morphology only
`lgbm_full`	both — the v1 model
V2 — feature engineering, physics, and the architecture we had recommended	
`lgbm_v2`	distances split by road class, cyclic hour
`physical`	line-source kernel, three positive parameters, *no morphology*
`hybrid`	physical kernel + LightGBM on the residual

Results

The ranking inverts — and that is the result



`models/metrics.json -- the same folds for every model`

Read across, not down

The elaborate architecture **wins the permissive split and loses both strict ones.** Three parameters is the only flat line.

What we did about it

The hybrid is **tested and rejected at this sample size** — not deferred to future work.

The delivered model

A line-source attenuation kernel. Two distances, three constants, **no morphology at all**:

$$E = \frac{A_{\mathrm{hw}}}{\max(d_{\mathrm{hw}}, D_0)} + \frac{A_{\mathrm{res}}}{\max(d_{\mathrm{res}}, D_0)} + B$$

$$L = 10 \log_{10} E$$

$A_{\mathrm{hw}} = 4.774 \times 10^7$ $A_{\mathrm{res}} = 3.800 \times 10^7$ $D_0 = 5\,\mathrm{m}$

- Positivity-constrained — every parameter is interpretable
- Ten lines of any language: no runtime, no server
- **B fits to $-99.6\,\mathrm{dB}$**, a boundary solution. **The background term is not identifiable here.**

`models/hybrid_physical.json` · `models/metrics.json`

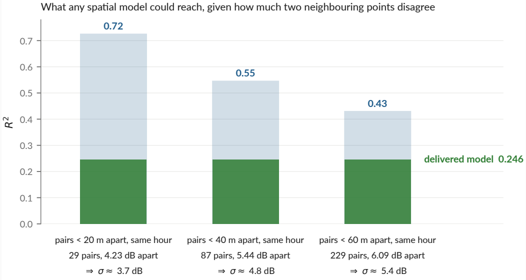
Performance under the reference protocol

R^2	**0.246**
95 % CI	[0.097, 0.339]
RMSE	6.20 dB
MAE	5.01 dB
data spread (σ)	7.14 dB

Say the error out loud

A typical prediction is wrong by **5 dB** — a factor of three in energy. Enough for a contrast, nowhere near an assessment.

“$R^2 = 0.25$ — isn’t that very low?”



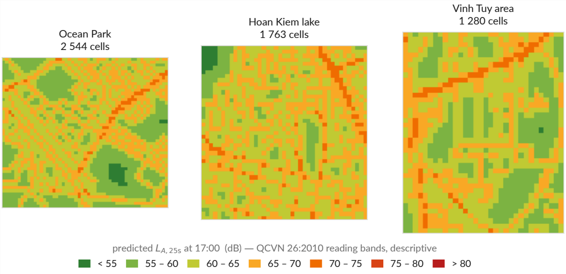
`data/processed/measurements.csv` · `docs/metrology.md`

Two points a few metres apart *at the same hour* must be predicted identically by any spatial model. Whatever they disagree by is variance **no spatial model can ever explain**.

At the 40 m scale the map works at

The ceiling is near $R^2 \approx 0.54$. The delivered model reaches **0.246** — about **half** of what is attainable, not a tenth of it.

The map — and the caveat that goes with it



`results/maps/hanoi_noise_map.csv -- 5 587 cells × 17 hours`

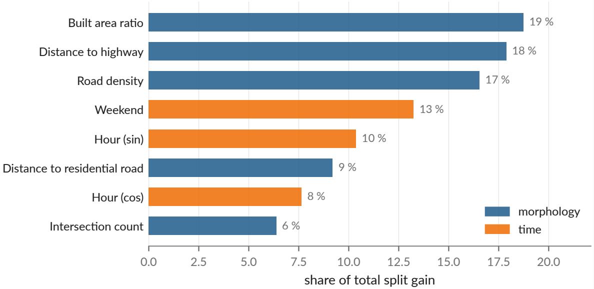
The grid step is not the accuracy

40 m cells, from a model whose typical error is **5 dB**. The bands are QCVN 26:2010, **descriptive**.

The envelope rule

Nothing is predicted where nothing was measured — `tests/test_grid_extent.py` enforces it.

What the learned model was leaning on



`models/metrics.json : feature_importance_gain`

Two thirds of LightGBM's split gain goes to **morphology** — built area, road density, distance to the two road classes.

And it buys nothing

Morphology alone scores $R^2 = \mathbf{0.041}$ under the reference protocol. **Where a model spends its capacity and where it earns its score are different questions**, and only the second one is a result.

This is why the ablation is on the same three splits as everything else.

The three negative results

1 — A three-parameter physical law beats every learned model we built

Six-variable gradient boosting, engineered features, a hybrid architecture: all lose the reference protocol to two distances and a constant. At $n = 363$ over three sites, there is not enough signal to justify capacity.

2 — Cross-city transfer fails, and in three distinct ways

Pretrained on Sunbird Uganda, scored on our 363 points: $R^2 = -15.9$ (v1), -8.2 (convention-invariant v2). **Not a calibration offset** — removing the 26 dB mean difference still leaves R^2 negative. **v1 transfers anti-information**, $r = -0.38$: it ranks quiet and loud the wrong way round. **v2 leaves nothing**, $r = +0.08$ — not wrong, empty. Against which $\log(d_{\mathrm{road}})$ alone reaches **0.240** on the same points.

3 — Vehicle flow does not explain the levels

Non-negative energy regression on 147 matched videos returns **zero coefficients** for motorcycles and cars. Speed and source–receiver distance are not observable from a non-georeferenced camera. **Section 4 revisits how much of this is physics and how much is instrument.**

`make transfer` · `scripts/experiments/uganda_transfer.py` · `docs/negative-results.md`

Auditing our own data

We found one of our own published results was wrong

Retracted, August 2026

A LightGBM model scored $R^2 = 0.45$ under what was then described as honest spatial cross-validation. A GAMA simulation was built on it. A project-update deck presented the figure.

The grouping was on **110 m cells** while the features aggregate over **300 m**. It leaked.

What replaced it

$R^2 = 0.246$ under buffered leave-one-out — roughly half the withdrawn figure, and the one that survives a split excluding the feature support.

- The retracted deck and map are in `docs/archive/`, **with their reason**, not deleted
- Nine corrections applied the same day
- The project pivoted from *producing a noise map* to *studying what this protocol supports*

`docs/audit/scientific-audit.md` · `docs/project-timeline.md` · `docs/archive/`

A second audit, on the data itself

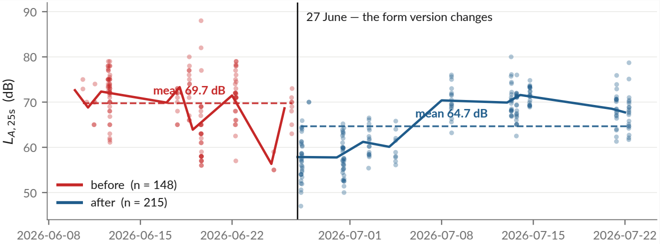
The first audit checked the *method*. This one checks the *measurements*.

One question, over the published CSVs, the raw Kobo export and the cleaning script: **does anything here behave in a way the documentation does not explain?**

Checked	Result	
Raw rows vs published rows	363 → 363, none dropped	**clean**
Every README figure vs `metrics.json`	exact match, none hand-copied	**clean**
Coordinate duplicates	none at 1 m	**clean**
Weather fields	consistent hourly API values	**clean**
Test suite	34 passing	**clean**
Level distribution over time	a discontinuity on 27 June	**finding**
Video→measurement chain	modal split implausible	**finding**
Confidence intervals in the headline table	overlapping	**finding**

The three findings are on the next two slides. None was known when the results section was written.

Finding 1 — a $-5.1\,\mathrm{dB}$ discontinuity on 27 June



`data/processed/measurements.csv` · `docs/audit/`

Controlling for site, hour and class: $-5.14\,\mathrm{dB}$ (SE 0.55, $p \approx 10^{-20}$). The form version changes the same day — integer readings 87 % $\rightarrow$ 20 %.

Why it matters

Before/after is predictable from $(\mathrm{lat}, \mathrm{lon})$ alone at **AUC 0.83**. An instrumental shift is then spatially structured, and a model can learn it **as morphology**. **Status:** **open**.

Findings 2 and 3 — the video chain, and the intervals

2 — The video chain is measuring something, but not what we think

- Detector modal split: **57 % cars, 36 % motorcycles**. In Hanoi. Reality is roughly the inverse.
- Video→measurement gaps: median 15 s, p90 56 s, **max 161 s** — for a 25 s sample
- Matched subset spans 61–80 dB against 47–88 dB overall: σ **cut by 42 %**
- Flow vs level: $r = -0.11$. A *negative* correlation between traffic and noise is not physical.

Consequence for negative result 3: it is far more likely an **instrument failure** than a finding about acoustics. It should be reported as *"the chain could not be made to work"*.

3 — The headline ranking is not statistically separated

`physical` $R^2 = 0.246$, CI $[0.097, 0.339]$ vs `dist_road` $R^2 = 0.200$, CI $[0.077, 0.266]$. The intervals overlap almost entirely. **"The physical kernel beats log-distance" is a coin flip, and the deck's own table should show the CIs.**

What the audit leaves standing

Holds

- The evaluation protocol and the ranking inversion
- Negative result 1 — capacity does not pay at $n = 363$
- Negative result 2 — transfer fails, $R^2 < 0$
- Provenance: raw $\rightarrow$ published is lossless and checkable
- Every published number traces to one file

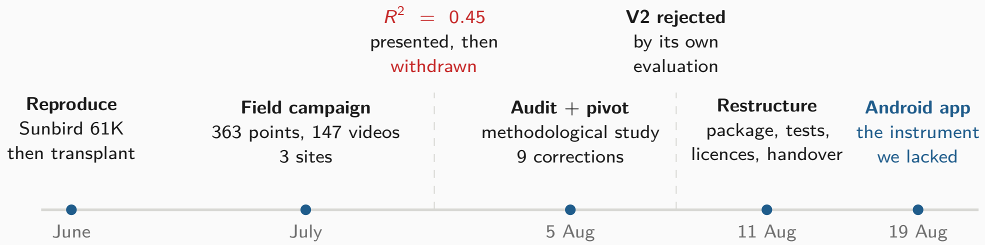
Needs work before submission

- The 27 June discontinuity — diagnose, then refit with the phase
- Negative result 3 — reframe as an instrument failure
- Carry the confidence intervals into every summary table
- Flag observed vs derived values in the published CSV (15 % of `construction_nearby` is inferred from the noise class)

The instrument was the problem in July, and it is still the problem in August.

Which is the argument for the next section.

From a pipeline to an instrument



The pattern

Every phase ended by **rejecting its own preferred answer**: transfer, then the leaked R^2, then the hybrid architecture we had recommended ourselves.

Where that led

Each rejection pointed at the same root cause — an instrument we did not control. So the last three weeks were spent building one.

`docs/project-timeline.md`

Why build an application at all

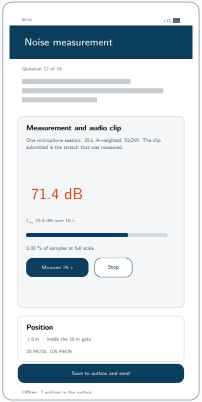
Four things the campaign could not do

1. **Control the instrument.** *Decibel X* is closed: no visible weighting, window or gain.
2. **Measure and record in one session.** Two apps meant the level of one stretch of street and the audio of another.
3. **Scale beyond three people.** Nobody pairs ODK Collect and a sound meter by hand.
4. **Record what the device was.** Handset, OS, audio source — none of it reached the dataset.

The decision, 19 August 2026

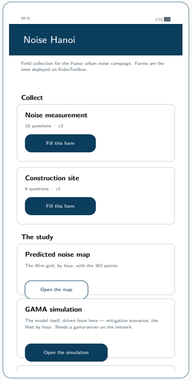
One application that **replaces the pairing** — forms, level, clip, GPS fix, submission, offline-first — and **carries the study**, negative results included.

`mobile/README.md` Kotlin + Compose, one `:app` module, `minSdk` 26



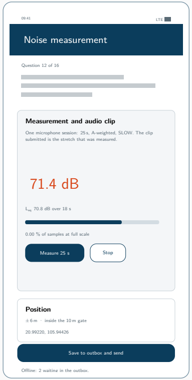
one session: level + clip

The application



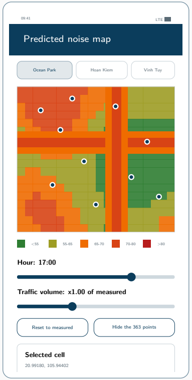
Collect

both forms, offline



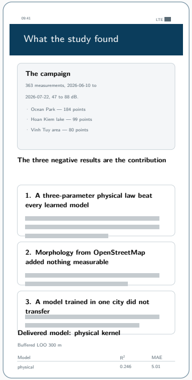
Measure

25 s, A-weighted



Map

40 m grid, by hour



Results

read from `metrics.json`

Vector mock-ups drawn from the Compose source. **No device capture exists yet** — which is the last slide of this section.

What was built — phase by phase

Phase	Goal	Delivered
1 — Collect	Replace ODK Collect	Both XLSForms as native Compose screens (16 + 6 questions), GPS with the 10 m gate, offline outbox drained by `WorkManager`, OpenRosa submission
2 — Measure	Replace Decibel X	`AudioRecord` at 48 kHz, A-weighting, SLOW, 25 s window after a discarded warm-up; AGC, noise suppression and echo cancellation **off**; the same PCM encoded to AAC for the clip
3 — Show	Carry the study	The 40 m grid by hour, the 363 points, the figures, and every metric *read* from `metrics.json` — negative results included
4 — Predict	On-device prediction	The delivered kernel in `study/Scenario.kt`, under test — and a **refusal** outside the three mapped areas
5 — Simulate	Reach the model	Tier 1 recomputed natively ($\times 0.2$ to $\times 3$, $k = 1$ asserted by a test) *and* the real model driven over `gama-server`
6 — Backend	Scale	**Not built.** Quota, moderation, deduplication, and the path back into `01_prepare_field_data.py`

5 278 lines of Kotlin, 33 files, 36 JVM unit tests `mobile/PLAN.md`

Architecture — and the one design rule that matters

`app/src/main/java/org/noisehanoi/mobile/`

`form/`	`the two XLSForms as data`
`odk/`	`instance XML, OpenRosa client`
`outbox/`	`on-disk queue + WorkManager`
`measure/`	`A-weighting, meter, PCM, AAC`
`location/`	`GPS fixes, the three sites`
`settings/`	`server config, mic offset`
`study/`	`grid, kernel, tier 1 scenario`
`ui/`	`Compose screens`

`form/`, `odk/`, `AWeighting.kt` and `PcmWriter.kt` are **pure Kotlin, no Android dependency** — which is why the constraint logic, the instance format, the weighting curve and the sample byte order are covered by JVM unit tests rather than by an emulator.

The rule: the app does not carry a copy of the study

`syncStudyData` in `app/build.gradle.kts` copies `hanoi_noise_map.csv`, `measurements.csv`, `metrics.json`, `hybrid_physical.json` and the figures **out of the repository at build time**.

Re-run the pipeline, rebuild the APK, and the app moves with it. No number is transcribed; nothing can drift.

Same principle, third time

The report reads `metrics.json`. The dashboard reads `metrics.json`. Now the app reads `metrics.json`. **Every published number has exactly one home.**

Three engineering results worth reporting

GAMA does not run on a phone — so the app reaches it two ways

Tier 1, recomputed natively over the 5 587 cells already exported: a traffic multiplier, offline, agreeing with the model to within $0.2\,\mathrm{dB}$. **The model itself, over `gama-server`**: the app draws GAMA's own display and drives it, which buys the mitigation scenarios the grid cannot hold — and needs a server the grid does not. **Drawing a model's display is not embedding it.**

The same correction, found a second time

The first on-device decomposition subtracted the kernel's constant ($-99.6\,\mathrm{dB}$, i.e. nothing) from a cell energy of order 10^6, so it scaled the total after all: at a fifth of the traffic, $-7.0\,\mathrm{dB}$ where the model gives -3.7. **Exactly the pedestrianisation error this project had already made and corrected**, rebuilt on another platform. A unit test now asserts that $k = 1$ moves nothing.

KoboToolbox renames your form at deployment

The first real submission failed with **404 for a form the server had never heard of**. Kobo discards the XLSForm's `id_string` on deploy and substitutes the asset's own identifier (`aA8FaTuUVSkRjbUW7rCBz7`). The app now reads `/formList` and submits under whatever the server calls the form. **None of this is visible from the phone** — the argument for testing against the real server rather than reasoning about it.

What the app measures — and what it refuses to claim

The app's level is a *different quantity* from the campaign's

This handset's microphone, not a trimmed Decibel X reading. It goes in **its own field**, with the device and the audio source the platform granted. **The two never share a column.**

Verified end to end, live

- Fix, 25 s measurement, instance written, **submission accepted by** `kc.kobotoolbox.org`
- A position outside the three areas is **refused, not answered**
- 36 unit tests; weighting pinned to the IEC table

Not verified — and it is the important one

The level has never been checked against a sound source. An emulator records silence. Until it is compared to a meter, **the absolute figure means nothing** — equally true, per `docs/metrology.md`, of the campaign's own numbers.

First task on a real handset, alongside Decibel X on the three campaign phones.

Next: Clermont-Ferrand

What Clermont-Ferrand unlocks that Hanoi could not

The binding constraint, restated

Every limitation in this study reduces to one fact: **no reference instrument was ever available**. Not borrowed, not rented, not bought.

What changes

At ISIMA / Université Clermont Auvergne, a **class 1 or class 2 sound level meter and a 94 dB / 1 kHz acoustic calibrator** are institutional equipment, not a budget line.

The three upgrades, in dependency order

1. **Anchor the absolute scale.** Calibrate the app against the reference meter across levels and handsets. Turn `micOffset` from a plausible constant into an **affine, measured correction**. Everything downstream inherits this.
2. **Validate the detector.** ∼10 videos, double-counted by hand, precision/recall/MAPE per class. *One day of work that turns an open-ended limitation into a quantified uncertainty.*
3. **Replace the kernel with real physics.** CNOSSOS-EU via **NoiseModelling**, corrected by a locally learned residual — the architecture the data has been pointing at since August.

`docs/handover.md` §6 "the scientifically indicated next step", debt #2, P0-3

The comparative deployment — and why it is the decisive test

	Hanoi	**Clermont-Ferrand**
Fleet	two-wheelers	cars, tram
Fabric	dense, organic	European, planned
Relief	flat delta	volcanic, hilly
Monitoring	none	EU END mapping

Held constant — the **same APK**, form, accuracy gate, $L_{A,25\mathrm{s}}$, OpenStreetMap features and evaluation protocols.

Why this is stronger than Uganda → Hanoi

There, the **instrument, protocol and feature conventions all differed too**: "transfer fails" and "the datasets are not commensurable" are not separable. One app in both cities and **the instrument stops being a confound**.

And a ground truth, finally

Clermont-Ferrand falls under EU END strategic mapping — the first **official, instrumented reference map** to compare against.

The programme, staged

Stage	Work	Gate — what must be true to proceed
S1 Instrument	Calibrate the app against a class 1/2 meter and a 94 dB calibrator, across levels and handsets. Affine correction, per device.	A reading from the app is traceable to a standard, with a stated uncertainty
S2 Repair	Diagnose the 27 June discontinuity and refit with the phase. Validate the detector. Carry the CIs everywhere.	The Hanoi dataset is defensible as published, or its defects are quantified
S3 Physics	CNOSSOS-EU via NoiseModelling + a learned residual, benchmarked against the delivered kernel on the same three splits.	Real propagation beats two distances and a constant — or it does not, and that is reportable too
S4 Deploy	Clermont-Ferrand campaign, same APK. Backend for scale (phase 6): quota, moderation, deduplication, path back into the pipeline.	A second dataset, commensurable with the first by construction
S5 Compare	Transfer both ways. Validate against EU END strategic mapping. Publish the two-city methodological study.	—

Sequenced by dependency, not by calendar. Framing, supervision and funding at Clermont-Ferrand are unconfirmed.

Conclusion

What this work established

Findings

1. At $n = 363$ over three sites, **three physical parameters beat every learned model** we built. Capacity does not pay.
2. **Cross-city transfer fails** — $R^2 < 0$, with matched features.
3. The video chain **could not be made to work**; that is an instrument result, not an acoustic one.
4. A cross-calibrated smartphone protocol supports **contrasts**, never levels.

Practices that made those findings trustworthy

- The buffer matches the feature radius — always
- Baselines and an ablation on *the same* splits
- The delivered model is chosen **by code**
- Every number has **exactly one home**
- Retracted work is archived **with its reason**
- Nothing is predicted outside the sampled envelope

The deliverable is a **protocol, an instrument and an honest error bar** — not a noise map of Hanoi.

Thank you

`github.com/Colauz/noise-modelling-hanoi`

Everything on these slides rebuilds from the repository:

`make features` `make models` `make results` `make report`

the deck's own figures: `scripts/12_presentation_figures.py`

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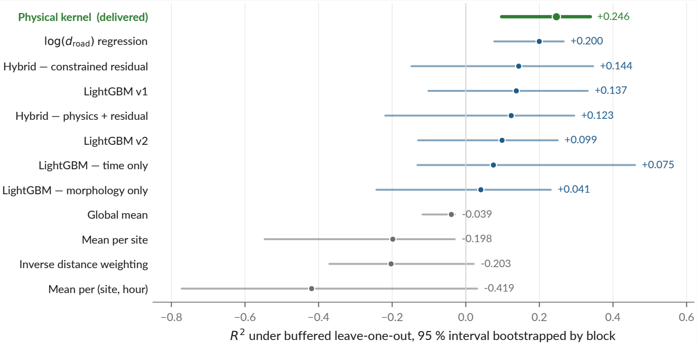
Backup — every model, with the interval that qualifies it



`models/metrics.json -- 2 000 resamples, bootstrapped by block`

This is audit finding 3, drawn

`physical` and `dist_road` overlap almost entirely. "The physical kernel beats log-distance" is a **coin flip** at this sample size. Other protocols: `models/model_comparison.md`.

Backup — the absolute-bias interval

Anchor	Instrument	Quantity	Raw gap	Corrected
Gelb & Apparicio 2019, HCMC	dosimeters + GPS	$L_{\mathrm{Aeq,1min}}$	+9.3	+7.3
Phan et al. 2010, Hanoi	RION NL-21/22	L_{den}	+7.0	+3.0
Phan et al. 2010, Hanoi	RION NL-21/22	L_{night}	−3.0	−3.0
IOHE, 12 arteries (press)	undocumented	daytime mean	+8.6	+7.6

Conclusion

Plausible absolute bias: $+3.0$ **to** $+7.3\,\mathrm{dB}$ (peer-reviewed sources, night excluded — $n = 10$ is too thin).
Corrected gap = raw gap minus the difference the quantity alone explains: it approximates the residual instrumental bias, and **is reported and never applied**.

`results/tables/literature_anchoring.md` · `scripts/06_anchor_literature.py`

Backup — what is published, and what is not

Dataset	Size	Published	Why / licence
Field measurements	363 pts, 3 sites	yes	CC-BY-4.0
Vehicle counts from video	147 videos	yes	CC-BY-4.0
Survey forms (XLSForm)	2 forms	yes	CC-BY-4.0
Traffic videos	147, 6.0 GB	no	faces and plates
Raw Kobo exports	—	no	collector identities
OpenStreetMap extract	49 146 buildings	no	regenerable; ODbL
Sunbird Uganda 61K	61 k clips	no	gated upstream

Stated plainly in the repository

No ethics review was requested for the video collection, and none is claimed. Only non-identifying aggregates are released. The retention decision is still **open**, and is a supervisor decision.

`docs/data-sources.md` · `LICENSE, LICENSE-DATA`